



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

MATHEMATICS
PAPER 2

9164/2

NOVEMBER 2016 SESSION

3 hours

Additional materials:
Answer paper
Graph paper
List of Formulae
Electronic calculator

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer all questions.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree and in other cases it should be given correct to 2 significant figures.

If a numerical value for g is necessary, take $g = 9.81 \text{ ms}^{-2}$.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

Within each section of the paper, questions are printed in the order of their mark allocations and candidates are advised, within each section, to attempt questions sequentially.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 6 printed pages and 2 blank pages.

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SECRETS THAT WILL MAKE MATHEMATICS EASY

Section A: Pure Mathematics

1 Differentiate with respect to x

(i) $x^2 \cos x,$ [2]

(ii) $\frac{e^{-x}}{x}.$ [2]

2 Prove by mathematical induction that

$$\sum_{r=1}^n 4^r = \frac{4}{3}(4^n - 1) \text{ for all positive integral values of } n.$$
 [7]

3 Show that $\sec 2x + \tan 2x = \tan\left(\frac{\pi}{4} + x\right).$ [7]

4 (i) Show by means of a graphical sketch or otherwise that the equation $e^x + 2x - 3 = 0$ has only one real root between 0 and 1.5 [3]

(ii) Taking 0.5 as the first approximation, use Newton-Raphson method twice to approximate the root of the equation $e^x + 2x - 3 = 0$ to 2 decimal places. [5]

5 The line l has equation $r = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + d \begin{pmatrix} 3 \\ 8 \\ -5 \end{pmatrix}$ and the plane π has equation

$$r \cdot \begin{pmatrix} -1 \\ 9 \\ 4 \end{pmatrix} = -20.$$

(i) Find the coordinates of R, the point of intersection of the line l and the plane π . [4]

(ii) Calculate the acute angle between the line l and plane π . [2]

(iii) Find the exact value of the perpendicular distance from a point Q with coordinates (1; 2; 3) to the plane π . [2]

6 (a) (i) Find the images of the points (2; 1) and (-2; 2) under a transformation with matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$ [3]

(ii) Describe fully the transformation in (i). [2]

(b) It is given that matrix $A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$ and that $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -5 & -2 & 1 \end{pmatrix}$ and

$$B^{-1} = \begin{pmatrix} 1 & -4 & 14 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}.$$

(i) Find $(AB)^{-1}$. [3]

(ii) Hence or otherwise, solve the following system of equations:

$$\begin{aligned} x &= 1 \\ x + y &= 2 \\ 3x + 2y + z &= 1 \end{aligned} \quad [3]$$

7 (i) Find the Binomial expansion of $\frac{1}{\sqrt{1-2x^2}}$ as far as the term in x^4 . [3]

(ii) State the range of values of x for which the expansion is valid. [1]

(iii) Integrate $\int_0^{0.5} \frac{1}{\sqrt{1-2x^2}} dx$, giving the answer correct to 2 decimal places. [4]

(iv) Use the Trapezium Rule with six ordinates to find the approximate integral of $\int_0^{0.5} \frac{1}{\sqrt{1-2x^2}} dx$. [4]

(v) Comment on the answers obtained in parts (iii) and (iv). [1]

8 (a) Using De Moivre's Theorem, show that $\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$. [6]

(b) Find the square roots of the complex number $3 + 4i$. [5]

(c) If the point P corresponds to the complex number z , shade on an Argand diagram the region defined by the inequalities $|z - z_1| \leq 3$ and $\frac{\pi}{6} \leq \arg(z - z_1) \leq \frac{\pi}{3}$ where $z_1 = 3 + 3i$. [3]

Section B: Mechanics

- 9 A block of weight 20 N is held at rest on a smooth plane inclined at 45° to the horizontal by a force of magnitude P N.

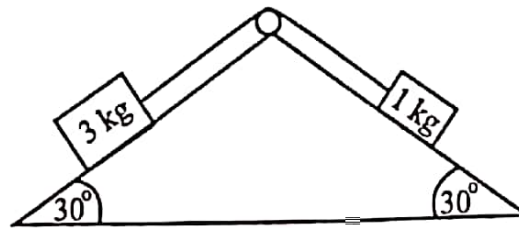
Find the value of P if it is

- (i) parallel to the plane,
(ii) applied horizontally.

[2]

[2]

10



Masses of 3 kg and 1 kg are held at rest on smooth inclined planes. The planes are inclined at angles of 30° to the horizontal. The masses are connected by a light inextensible string passing over a smooth fixed pulley (see diagram). The system is then released from rest.

Find the

- (i) acceleration of the system,
(ii) tension in the string.

[3]

[2]

- 11 An overloaded minibus is travelling at a constant speed of 90 km/h when it passes a stationary police car on the road side. The police car immediately gives chase, accelerating uniformly to reach a speed of 120 km/h in 10 seconds and continues at this speed until it overtakes the minibus.

Find the

- (i) acceleration of the police car during the first 10 seconds,
(ii) time taken by the police car to catch up with the minibus,
(iii) distance travelled by the police car to catch up with the minibus.

[2]

[2]

[2]

- 12 A particle is projected horizontally from a point 100 m above a horizontal plane with a speed of 25 ms^{-1} .

Find the

- (i) speed and direction of the particle after 2 seconds, [5]
- (ii) time taken by the particle to reach the horizontal plane, [2]
- (iii) horizontal displacement of the particle from point of projection to the point it reaches the horizontal plane. [2]

Section C: Statistics

- 13 The random variable X has a probability density function given by

$$f(x) = \begin{cases} 2x^k & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{where } k \text{ is an integer.}$$

Find the

- (a) value of k , [2]
 (b) expected value of X . [2]

- ✓14 In a national election of a country, 65% of the voters population support Party A, 30% support Party B and 5% support Party C. A referendum in the same country on an issue is to be decided by a "Yes" or a "No" vote. A survey carried out indicated that 48% of the voters population would vote "Yes".

- (a) Find the probability that a voter votes "No" and supports Party A. [2]
 (b) Given that a voter voted "No", calculate the probability that the voter is from Party A. [3]

- * 15 The marks obtained by candidates in an examination, marked out of 100, are given in the table.

mark	≤ 44	45-54	55-59	60-74	≥ 75
frequency	90	70	75	45	25

- (a) The height of the bar of a histogram of the class 55-59 is 15.
 Calculate the heights of the other bars of the histogram. [3]
 (b) Hence construct a histogram to represent this data. [2]

- 16 The probability that a learner passes a driving test is 0.24. Ten learners from a driving school took the test.

Calculate the probability that

- (a) (i) exactly three learners passed the test, [3]
 (ii) more than eight learners failed the test. [3]
 (b) Another driving school sent 45 learners for the test.

Calculate the probability that at most 18 learners passed the driving test. [4]